

Experiment 3: Creation of a Simple Network Topology Using Open-Source Network Virtualization Tools

Objective

To create and configure a simple network topology using an open-source network virtualization tool and verify communication between the virtual network nodes.

Task

- Ubuntu Linux Virtual Machine
- VirtualBox
- Mininet
- Create Network Topology on each type with 2 different type of node numbers.
- See links and ping with basic Linux networking commands
- After completion, stop the network.

1) Installation of Mininet :-

a. First Command :-

sudo apt update

b. Second command :-

sudo apt install mininet

```
Sep 21 17:19
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~

sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [480 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [616 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [113 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [294 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [403 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [78.3 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1,056 B]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [173 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Fetched 2,945 kB in 4s (657 kB/s)
141 packages can be upgraded. Run 'apt list --upgradable' to see them.
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ sudo apt install mininet
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Installing:
  mininet

Installing dependencies:
  cgroup-tools  libcgroup3  libunbound8  net-tools  openvswitch-switch  python3-openvswitch  python3-sortedcontainers
  iperf        libevent-2.1-7t64  libxdp1  openvswitch-common  python3-netifaces  python3-packaging  socat

Suggested packages:
  openvswitch-doc python3-graphviz python3-unbound python-sortedcontainers-doc

Summary:
  Upgrading: 0, Installing: 15, Removing: 0, Not Upgrading: 141
```

```
Sep 21 17:20
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~

sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [173 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Fetched 2,945 kB in 4s (657 kB/s)
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  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Installing:
  mininet

Installing dependencies:
  cgroup-tools  libcgroup3  libunbound8  net-tools  openvswitch-switch  python3-openvswitch  python3-sortedcontainers
  iperf        libevent-2.1-7t64  libxdp1  openvswitch-common  python3-netifaces  python3-packaging  socat

Suggested packages:
  openvswitch-doc python3-graphviz python3-unbound python-sortedcontainers-doc

Summary:
  Upgrading: 0, Installing: 15, Removing: 0, Not Upgrading: 141
  Download size: 5,625 kB
  Space needed: 19.2 MB / 13.3 GB available

Continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libcgroup3 amd64 3.1.0-2build1 [63.0 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 cgroup-tools amd64 3.1.0-2build1 [85.6 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 iperf amd64 2.2.1+dfsg-1build1 [161 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libevent-2.1-7t64 amd64 2.1.12-stable-10ubuntu0.1 [145 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libunbound8 amd64 1.24.2-1ubuntu2.2 [456 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libxdp1 amd64 1.6.2-1ubuntu1 [65.7 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 net-tools amd64 2.10-2ubuntu1 [151 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 python3-netifaces amd64 0.11.0-2build7 [17.9 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 python3-packaging all 26.0-1 [58.7 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 python3-sortedcontainers all 2.4.0-2build1 [27.1 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 socat amd64 1.8.1.1-1ubuntu0.1 [403 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 mininet amd64 2.3.0-1.1build1 [131 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 python3-netifaces amd64 0.11.0-2build7 [17.9 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 python3-packaging all 26.0-1 [58.7 kB]
```

```
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sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

Download size: 5,625 kB
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Continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 libcgroup3 amd64 3.1.0-2build1 [63.0 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 cgroup-tools amd64 3.1.0-2build1 [85.6 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 iperf amd64 2.2.1+dfsg-1build1 [161 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libevent-2.1-7t64 amd64 2.1.12-stable-10ubuntu0.1 [145 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libunbound8 amd64 1.24.2-1ubuntu2.2 [456 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libxdp1 amd64 1.6.2-1ubuntu1 [65.7 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 net-tools amd64 2.10-2ubuntu1 [151 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 python3-netifaces amd64 0.11.0-2build7 [17.9 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 python3-packaging all 26.0-1 [58.7 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 python3-sortedcontainers all 2.4.0-2build1 [27.1 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 socat amd64 1.8.1.1-1ubuntu0.1 [483 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 mininet amd64 2.3.0-1.1build1 [131 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 openvswitch-common amd64 3.7.1-2 [1,081 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 openvswitch-switch amd64 3.7.1-2 [1,683 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 python3-openvswitch amd64 3.7.1-2 [1,097 kB]
Fetched 5,625 kB in 9s (593 kB/s)
Selecting previously unselected package libcgroup3:amd64.
(Reading database... 195749 files and directories currently installed.)
Preparing to unpack ../00-libcgroup3_3.1.0-2build1_amd64.deb...
Unpacking libcgroup3:amd64 (3.1.0-2build1)...
Selecting previously unselected package cgroup-tools.
Preparing to unpack ../01-cgroup-tools_3.1.0-2build1_amd64.deb...
Unpacking cgroup-tools (3.1.0-2build1)...
Selecting previously unselected package iperf.
Preparing to unpack ../02-iperf_2.2.1+dfsg-1build1_amd64.deb...
Unpacking iperf (2.2.1+dfsg-1build1)...
Selecting previously unselected package libevent-2.1-7t64:amd64.
Preparing to unpack ../03-libevent-2.1-7t64_2.1.12-stable-10ubuntu0.1_amd64.deb...
Unpacking libevent-2.1-7t64:amd64 (2.1.12-stable-10ubuntu0.1)...
Selecting previously unselected package libunbound8:amd64.
Preparing to unpack ../04-libunbound8_1.24.2-1ubuntu2.2_amd64.deb...
Unpacking libunbound8:amd64 (1.24.2-1ubuntu2.2)...
Selecting previously unselected package libxdp1:amd64.
Preparing to unpack ../05-libxdp1_1.6.2-1ubuntu1_amd64.deb...
```

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Sep 21 17:20
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

Preparing to unpack ../01-cgroup-tools_3.1.0-2build1_amd64.deb...
Unpacking cgroup-tools (3.1.0-2build1)...
Selecting previously unselected package iperf.
Preparing to unpack ../02-iperf_2.2.1+dfsg-1build1_amd64.deb...
Unpacking iperf (2.2.1+dfsg-1build1)...
Selecting previously unselected package libevent-2.1-7t64:amd64.
Preparing to unpack ../03-libevent-2.1-7t64_2.1.12-stable-10ubuntu0.1_amd64.deb...
Unpacking libevent-2.1-7t64:amd64 (2.1.12-stable-10ubuntu0.1)...
Selecting previously unselected package libunbound8:amd64.
Preparing to unpack ../04-libunbound8_1.24.2-1ubuntu2.2_amd64.deb...
Unpacking libunbound8:amd64 (1.24.2-1ubuntu2.2)...
Selecting previously unselected package libxdp1:amd64.
Preparing to unpack ../05-libxdp1_1.6.2-1ubuntu1_amd64.deb...
Unpacking libxdp1:amd64 (1.6.2-1ubuntu1)...
Selecting previously unselected package net-tools.
Preparing to unpack ../06-net-tools_2.10-2ubuntu1_amd64.deb...
Unpacking net-tools (2.10-2ubuntu1)...
Selecting previously unselected package python3-netifaces:amd64.
Preparing to unpack ../07-python3-netifaces_0.11.0-2build7_amd64.deb...
Unpacking python3-netifaces:amd64 (0.11.0-2build7)...
Selecting previously unselected package python3-packaging.
Preparing to unpack ../08-python3-packaging_26.0-1_all.deb...
Unpacking python3-packaging (26.0-1)...
Selecting previously unselected package python3-sortedcontainers.
Preparing to unpack ../09-python3-sortedcontainers_2.4.0-2build1_all.deb...
Unpacking python3-sortedcontainers (2.4.0-2build1)...
Selecting previously unselected package socat.
Preparing to unpack ../10-socat_1.8.1.1-1ubuntu0.1_amd64.deb...
Unpacking socat (1.8.1.1-1ubuntu0.1)...
Selecting previously unselected package mininet.
Preparing to unpack ../11-mininet_2.3.0-1.1build1_amd64.deb...
Unpacking mininet (2.3.0-1.1build1)...
Selecting previously unselected package openvswitch-common.
Preparing to unpack ../12-openvswitch-common_3.7.1-2_amd64.deb...
Unpacking openvswitch-common (3.7.1-2)...
Selecting previously unselected package openvswitch-switch.
Preparing to unpack ../13-openvswitch-switch_3.7.1-2_amd64.deb...
Unpacking openvswitch-switch (3.7.1-2)...
```

```
Sep 21 17:20
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

Unpacking socat (1.8.1.1-1ubuntu0.1)...
Selecting previously unselected package mininet.
Preparing to unpack ../11-mininet_2.3.0-1.1build1_amd64.deb...
Unpacking mininet (2.3.0-1.1build1)...
Selecting previously unselected package openvswitch-common.
Preparing to unpack ../12-openvswitch-common_3.7.1-2_amd64.deb...
Unpacking openvswitch-common (3.7.1-2)...
Selecting previously unselected package openvswitch-switch.
Preparing to unpack ../13-openvswitch-switch_3.7.1-2_amd64.deb...
Unpacking openvswitch-switch (3.7.1-2)...
Selecting previously unselected package python3-openvswitch.
Preparing to unpack ../14-python3-openvswitch_3.7.1-2_amd64.deb...
Unpacking python3-openvswitch (3.7.1-2)...
Setting up net-tools (2.10-2ubuntu1)...
Setting up libcgroup3:amd64 (3.1.0-2build1)...
Setting up libevent-2.1-7t64:amd64 (2.1.12-stable-10ubuntu0.1)...
Setting up cgroup-tools (3.1.0-2build1)...
Setting up python3-sortedcontainers (2.4.0-2build1)...
Setting up libxdpi:amd64 (1.6.2-1ubuntu1)...
Setting up python3-packaging (26.0-1)...
Setting up libunbound8:amd64 (1.24.2-1ubuntu2.2)...
Setting up lperf (2.2.1+dfsg-1build1)...
Setting up socat (1.8.1.1-1ubuntu0.1)...
Setting up python3-netifaces:amd64 (0.11.0-2build7)...
Setting up openvswitch-common (3.7.1-2)...
Setting up mininet (2.3.0-1.1build1)...
Setting up openvswitch-switch (3.7.1-2)...
update-alternatives: using /usr/lib/openvswitch-switch/ovs-vswitchd to provide /usr/sbin/ovs-vswitchd (ovs-vswitchd) in auto mode
Created symlink '/etc/systemd/system/openvswitch-switch.service.requires/ovs-record-hostname.service' → '/usr/lib/systemd/system/ovs-record-hostname.s
ervice'.
Created symlink '/etc/systemd/system/openvswitch.service' → '/usr/lib/systemd/system/openvswitch-switch.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/openvswitch-switch.service' → '/usr/lib/systemd/system/openvswitch-switch.service'.
Setting up python3-openvswitch (3.7.1-2)...
Processing triggers for libc-bin (2.43-2ubuntu2.4)...
Processing triggers for ufw (0.36.2-9build1)...
Processing triggers for man-db (2.13.1-1build1)...
sagnik-bachhar@sagnik-bachhar-VirtualBox: ~$
```

2) Verify the installation of Mininet & Check The Version Of Mininet :-

```
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ mn --version
2.3.0
sagnik-bachhar@sagnik-bachhar-VirtualBox:~$
```